Lab Experiment-16

Aim

To write a Python program using the spaCy library to identify and extract named entities such as names of people, organizations, locations, dates, and other entities from a given text.

Algorithm

1. Import the spaCy library.
2. Load the pre-trained English language model.
3. Give a text as input.
4. Process the text using the spaCy NLP model.
5. Extract the named entities using doc.ents.
6. Display each entity and its entity label.
7. Display the result.

Python Code

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = input("Enter a sentence: ")

doc = nlp(text)

print("\nNamed Entities:")

for ent in doc.ents:

    print(ent.text, "-", ent.label_)
```

Output:

```python
import spacy

# Load English language model
nlp = spacy.load("en_core_web_sm")

# Input text
text = "Swarna studies at SIMATS in Chennai. She joined the college in 2024."

# Process the text
doc = nlp(text)

# Display named entities
print("Named Entities:")

for ent in doc.ents:
    print(ent.text, "-", ent.label_)
```

```
Named Entities:
Swarna - GPE
SIMATS - ORG
Chennai - GPE
2024 - DATE
```

Result

Thus, the Python program successfully performs Named Entity Recognition (NER) using the spaCy library and identifies entities from the given text.